Class 9 Mathematics — Exercise 4.5

Basic Concept

Relation between HCF and LCM

For two polynomials P and Q:

$$\boxed{\text{LCM} \times \text{HCF} = P \times Q}$$

So:

$$\text{LCM} = \frac{P \times Q}{\text{HCF}} \qquad \text{HCF} = \frac{P \times Q}{\text{LCM}} \qquad Q = \frac{\text{LCM} \times \text{HCF}}{P}$$

Reminder: HCF = common factors with least power; LCM = Common factors $\times$ Uncommon factors.

Question 1

Find the HCF and LCM of the following.

(i) $16 - 4x^2, \quad x^2 + x - 6$

For $x^2 + x - 6$: $-6 \to 3$ and $-2 \quad (3 - 2 = 1)$

$16 - 4x^2$	$x^2 + x - 6$
$= 4[4 - x^2]$	$= x^2 + 3x - 2x - 6$
$= 4[(2)^2 - (x)^2]$	$= x(x + 3) - 2(x + 3)$
$= 4(2 - x)(2 + x)$	$= (x - 2)(x + 3)$
$= -4(x - 2)(x + 2)$	

$$\text{H.C.F} = \boxed{x - 2}$$

$$\text{L.C.M} = (x - 2) \times (-4)(x + 2)(x + 3) = \boxed{-4(x - 2)(x + 2)(x + 3)}$$

(ii) $a^4 - a^3 - a + 1, \quad a^4 + a^2 + 1$

First:

$$a^4 - a^3 - a + 1 = a^3(a - 1) - 1(a - 1)$$
$$= (a - 1)(a^3 - 1)$$

$$= (a-1)[(a)^3 - (1)^3]$$
$$= (a-1)(a-1)[(a)^2 + (a)(1) + (1)^2]$$
$$= (a-1)(a-1)(a^2 + a + 1)$$

Second:

$$a^4 + a^2 + 1 = (a^2)^2 + (1)^2 + 2(a^2)(1) - 2(a^2)(1) + a^2$$
$$= (a^2+1)^2 - 2a^2 + a^2$$
$$= (a^2+1)^2 - (a)^2$$
$$= (a^2 + 1 + a)(a^2 + 1 - a)$$
$$= (a^2 + a + 1)(a^2 - a + 1)$$

$$\text{H.C.F} = \boxed{a^2 + a + 1}$$

$$\text{L.C.M} = (a^2 + a + 1) \times (a-1)(a-1)(a^2 - a + 1) = \boxed{(a^2 + a + 1)(a-1)^2(a^2 - a + 1)}$$

(iii) $x^3 + 2x^2 - 3x, \quad 2x^3 + 5x^2 - 3x$

$x^3 + 2x^2 - 3x$	$2x^3 + 5x^2 - 3x$
$= x(x^2 + 2x - 3)$	$= x(2x^2 + 5x - 3)$
$= x[x^2 + 3x - x - 3]$	$= x[2x^2 + 6x - x - 3]$
$= x[x(x+3) - 1(x+3)]$	$= x[2x(x+3) - 1(x+3)]$
$= x(x+3)(x-1)$	$= x(x+3)(2x-1)$

$$\text{H.C.F} = \boxed{x(x+3)}$$

$$\text{L.C.M} = x(x+3) \times (x-1)(2x-1) = \boxed{x(x+3)(x-1)(2x-1)}$$

Question 2

HCF and LCM of two polynomials are $x - 7$ and $x^3 - 10x^2 + 11x + 70$ respectively. Find the product of the two polynomials.

$$\text{HCF} = x - 7, \qquad \text{LCM} = x^3 - 10x^2 + 11x + 70, \qquad P \times Q = ?$$

($\because \text{LCM} \times \text{HCF} = P \times Q$)

$$P \times Q = \text{HCF} \times \text{LCM}$$
$$= (x-7)(x^3 - 10x^2 + 11x + 70)$$
$$= x(x^3 - 10x^2 + 11x + 70) - 7(x^3 - 10x^2 + 11x + 70)$$
$$= x^4 - 10x^3 + 11x^2 + 70x - 7x^3 + 70x^2 - 77x - 490$$
$$= \boxed{x^4 - 17x^3 + 81x^2 - 7x - 490} \text{ Ans}$$

Question 3

The HCF of two polynomials is $x-2$ and their product is $x^4+3x^3-12x^2-20x+48$. Find their LCM.

$$\text{HCF} = x-2, \qquad \text{LCM} = ?$$

$$\text{LCM} \times \text{HCF} = P \times Q$$
$$\text{LCM} = \frac{P \times Q}{\text{HCF}} = \frac{x^4+3x^3-12x^2-20x+48}{x-2}$$

Division:

$$x^4+3x^3-12x^2-20x+48-x^3(x-2) = 5x^3-12x^2-20x+48$$
$$5x^3-12x^2-20x+48-5x^2(x-2) = -2x^2-20x+48$$
$$-2x^2-20x+48-(-2x)(x-2) = -24x+48$$
$$-24x+48-(-24)(x-2) = 0$$

$\text{Quotient} = x^3+5x^2-2x-24, \qquad \text{Remainder} = 0$

$$\text{LCM} = \boxed{x^3+5x^2-2x-24} \text{ Ans}$$

Question 4

The LCM of two polynomials is $y^3+2y^2-11y-12$ and their product is $y^4+6y^3-3y^2-56y-48$. Find their HCF.

$$\text{LCM} \times \text{HCF} = P \times Q$$
$$\text{HCF} = \frac{P \times Q}{\text{LCM}} = \frac{y^4+6y^3-3y^2-56y-48}{y^3+2y^2-11y-12}$$

Division:

$$y^4+6y^3-3y^2-56y-48-y(y^3+2y^2-11y-12) = 4y^3+8y^2-44y-48$$
$$4y^3+8y^2-44y-48-4(y^3+2y^2-11y-12) = 0$$

$\text{Quotient} = y+4, \qquad \text{Remainder} = 0$

$$\text{HCF} = \boxed{y+4} \text{ Ans}$$

Question 5

Find the second polynomial when the first polynomial $= x^4+x^3+x+1$, HCF $= x+1$ and LCM $= (x^3+1)(x^4+x^3-x-1)$.

$$P \times Q = \text{LCM} \times \text{HCF}$$
$$Q = \frac{\text{LCM} \times \text{HCF}}{P} = \frac{(x^3+1)(x^4+x^3-x-1)(x+1)}{x^4+x^3+x+1}$$

Factorize the denominator:

$$x^4 + x^3 + x + 1 = x^3(x+1) + 1(x+1) = (x+1)(x^3+1)$$

$$Q = \frac{(x^3+1)(x^4+x^3-x-1)(x+1)}{(x+1)(x^3+1)}$$

$$Q = \boxed{x^4 + x^3 - x - 1} \text{ Ans}$$

Question 6

Find the LCM of the polynomials $4x^3 - 10x^2 + 4x + 2$ and $3x^4 - 2x^3 - 3x + 2$ if their HCF is $x - 1$.

$$P = 4x^3 - 10x^2 + 4x + 2, \qquad Q = 3x^4 - 2x^3 - 3x + 2, \qquad \text{HCF} = x - 1$$

$$\text{LCM} \times \text{HCF} = P \times Q$$

$$\text{LCM} = \frac{P \times Q}{\text{HCF}} = \frac{(4x^3 - 10x^2 + 4x + 2)(3x^4 - 2x^3 - 3x + 2)}{x - 1}$$

Divide $3x^4 - 2x^3 - 3x + 2$ by $x - 1$:

$$3x^4 - 2x^3 - 3x + 2 - 3x^3(x-1) = x^3 - 3x + 2$$
$$x^3 - 3x + 2 - x^2(x-1) = x^2 - 3x + 2$$
$$x^2 - 3x + 2 - x(x-1) = -2x + 2$$
$$-2x + 2 - (-2)(x-1) = 0$$

Quotient $= 3x^3 + x^2 + x - 2$, Remainder $= 0$

$$\text{LCM} = \boxed{(4x^3 - 10x^2 + 4x + 2)(3x^3 + x^2 + x - 2)} \text{ Ans}$$